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CORDIS Results Pack on AI in life sciences

Harnessing the transformative power of AI for scientific discovery

A thematic collection of innovative EU-funded research results

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Editorial

Harnessing the transformative power of AI for scientific discovery

AI techniques are helping researchers to make valuable new discoveries across disciplines such as biology, neuroscience and biotechnology. Through the analysis of vast datasets and the identification of complex patterns, AI is speeding up the discovery of new therapeutic drugs and helping us to monitor and manage our resources better. This CORDIS Results Pack showcases the application of AI in 15 EU-funded projects in this field, providing new insights in life sciences.

The disruptive potential of AI is only just beginning to be realised. Data analyses that might once have taken weeks, months or even years can now be performed far quicker, leading to faster results. Research into life sciences – an area that studies all life on our planet, from microorganisms, plants, animals and humans to whole ecosystems – is primed to take advantage of this technological revolution.

For example, machine learning algorithms capable of analysing complex biological data are increasingly being used to make predictive health models, opening the door to more personalised healthcare. AI tools can also help to simulate the potential effect of novel treatments and drugs, identifying promising molecules and highlighting potential side effects well before clinical trials begin.

It is notable that the 2024 Nobel Prizes in Physics and Chemistry were essentially for AI in science, with the Nobel Prize in Chemistry being awarded for AI research into protein structure prediction. "I've always thought if we could build AI in the right way, it could be the ultimate tool to help scientists," says DeepMind cofounder and Nobel Laureate Demis Hassabis.

Beyond the fields of health and medicine, the ability to analyse vast amounts of disparate information from a range of sources can significantly improve decision-making processes. Policymakers trying to protect ecosystems for example need to take account of a range of complex issues before identifying the best path forward. AI-powered tools can help to collate and structure such information in a usable and practical manner.

Boosting innovation and competitiveness

This Results Pack presents a diverse portfolio of projects supported by the European Union, which demonstrate how the application of AI is facilitating new discoveries in life sciences. AI in science in general, and life sciences in particular, are among the political priorities of the European Commission, as highlighted by President Ursula von der Leyen's Mission Letter to the Commissioner for Start-ups, Research and Innovation.

In the letter, President von der Leyen calls for "a new multi-disciplinary Strategy for European Life Sciences, to unlock high-value technologies in support of green and digital transitions", as well as for "a strategy to increase the uptake of AI by European scientists". These new strategies will be published in 2025.

The projects highlighted in this Results Pack support the European Union's work in this field and will help to maintain the EU's position as a global pioneer in science and in AI. This will be critical to boosting European competitiveness and the creation of new sustainable jobs.

AI-driven approaches to personalised healthcare

Applying AI to datasets covering health as well as other risk factors could be key to developing more predictive and personalised healthcare.



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Identifying individuals at high risk of disease quickly and efficiently is critical to achieving better long-term health outcomes. This is especially true for treatable conditions such as diabetes, stroke and coronary artery disease.

The AI-PREVENT project, funded by the European Research Council, sought to support this goal through the application of AI. For this purpose, health data of the Finnish population, along with a variety of other information, was gathered.

"This data was brought together through the FinRegistry initiative," explains AI-PREVENT project coordinator Andrea Ganna from the University of Helsinki. "This dataset includes health data as well as a wide range of other information such as hospitalisations, diagnoses, familial risk and socio-demographic status."

Modelling health and other risk factors

This valuable data source enabled the project to develop and apply novel AI approaches to model possible health trajectories with regards to cardiovascular disease. The aim was to find new ways of modelling the complex relationships between health and various other risk factors.

Applying AI tools to a large data registry means we can screen people better.

"You can think of this dataset as the book of life of all Finnish citizens," says Ganna. "Each individual is a sentence in this book. All these sentences are connected, with relationships and multigenerational elements."

Applying AI tools enabled Ganna and his team to examine each sentence in its context, and to generate predictions for the future. "We can try to see how these individual stories are likely to evolve," he explains.

Linking socio-economic data with health data

The AI-generated models were indeed shown to be powerful tools for identifying specific individuals at risk and in need of preventive care. Likely end uses could be to screen for cancer, or to identify individuals in need of a cardiovascular check-up.

"This kind of sorting would normally just be done on key criteria like age and gender," adds Ganna. "Applying AI tools to a large data registry means we can screen people better. This more personalised approach could help people live longer and be more cost-effective for health services."

The project also looked at the concept of fairness. Ganna wanted to get an idea of whether AI modelling would be accurate across all groups of society, or whether it might favour certain groups such as the well-off.

"We wanted to see whether linking socio-economic data with health data can deliver unbiased indicators," he notes. "This approach also opens up the question of whether, as a society, we even want to use data in this way. If an AI system is deciding who should be first to receive a vaccine based on complex data, will citizens accept this?"

Making AI models more generalisable

An important next step will be to expand this exercise across several European countries. Ganna notes that the European Health Data Space Regulation, which came into force in March 2025, is very much in line with legislation in Finland that made the AI-PREVENT project possible.

"This project has underlined the importance of data access," he says. "This is the core of everything we do. We won't be able to understand the fairness of applying AI tools, for example, unless we have actual data we can use. This research into precision public health doesn't happen without this foundational work."

Applying AI to more EU countries would help to make Ganna's models more generalisable, he believes, and capable of understanding differences between countries. He also intends to train his AI models on genetic data, to further strengthen preventive healthcare in Europe. These targeted approaches should help to improve health outcomes, and make lab work more effective and efficient.

PROJECT
AI-PREVENT – A nationwide AI risk assessment for primary prevention of cardiometabolic diseases

COORDINATED BY
University of Helsinki in Finland

FUNDED UNDER
Horizon 2020-ERC

CORDIS FACTSHEET
cordis.europa.eu/project/id/945733

PROJECT WEBSITE
finregistry.fi

AI brings healthcare to the edge of tomorrow

Tools developed by the AI-SPRINT project are helping healthcare researchers to integrate the power of data into patient care.



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AI has the potential to have a major impact on the life sciences, especially healthcare. But delivering on that potential will require data – and lots of it.

To get that data, developers must be willing to go to the edge.

"AI's benefits will largely come from using IT resources found at the periphery of the network, where the source of the data is produced," explains Danilo Ardagna, an associate professor at the Polytechnic University of Milan.

This is where edge computing comes into play. Unlike traditional cloud computing, where data is sent to a remote data centre for processing, edge computing is a distributed framework that brings processing power and data storage closer to the source of the data – typically near the end devices that generate and consume it.

"The advantage of processing data locally is that it reduces latency, improves application performance and enables real-time insights," says Ardagna.

Supporting the development of AI applications

Helping companies and researchers fully leverage these benefits is the EU-funded AI-SPRINT project. "Our goal is to democratise access to AI technologies by simplifying and accelerating the development of AI applications through edge computing," adds Ardagna, who coordinated the project.

AI and edge computing will revolutionise healthcare research and patient care.

At the heart of the project is a suite of novel tools designed to support the development of AI applications, along with securing their execution, ensuring easy deployment and optimising runtime management.

"These tools are unique in that they allow for the performance, energy efficiency and accuracy of the AI models while simultaneously providing security and privacy guarantees," notes Ardagna.

AI tools in the life sciences

Beyond designing the AI tools, the project also applied these tools to several use cases – including in the life sciences. For example, the Life Sciences Department of the Barcelona Supercomputing Center (BSC) integrated AI-SPRINT's technology to craft AI models for stroke risk assessment using sensor data and lifestyle information.

Specifically, BSC used AI-SPRINT solutions to safeguard sensitive healthcare data and optimise its risk forecasting models. "Our unique framework not only enables the incorporation of wearable and mobile devices into healthcare but ensures that these devices can leverage AI to provide valuable insights into patient care," remarks Ardagna.

The BSC personalised healthcare use case was so successful that a spin-off company was established to promote the use of its stroke risk assessment application. The company has since raised nearly EUR 700 000 in funding.

BSC also partnered with the Sant Pau Stroke Research Centre in Barcelona to develop a more advanced solution for risk prevention that includes multi-omics data.

AI and edge computing revolutionising healthcare

An unexpected outcome of the project was BSC's use of the AI-SPRINT models to acquire electrocardiogram (ECG) data from patients participating in a study conducted by the Ospedale Maggiore di Niguarda in Milan.

Following the completion of the study, the participating clinicians identified a patient with significant cardiovascular and heart-related concerns. This timely intervention, made possible thanks to the use of AI models, allowed for the commencement of additional analyses and treatment, thereby mitigating potential health complications.

"This is but one example of how AI and edge computing will revolutionise healthcare research and patient care – and AI-SPRINT is proud to have contributed some of the tools that are making this revolution possible," concludes Ardagna.

PROJECT
AI-SPRINT – AI in Secure PRIvacy-preserving computing coNTinuum

COORDINATED BY
Polytechnic University of Milan in Italy

FUNDED UNDER
Horizon 2020-LEIT-ICT

CORDIS FACTSHEET
cordis.europa.eu/project/id/101016577

PROJECT WEBSITE
ai-sprint-project.eu

Cutting-edge techniques to address spinal degeneration

AI and computer simulation-driven tools can provide critical clinical support to identify and address spinal degeneration.



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While clinical observations provide insights into people's health, converting such evidence into well-informed disease prevention and treatment remains challenging. This is the case in spine degeneration and low back pain, where tissue and organ degradation that lead to clinical symptoms are often highly multifactorial.

"The complex interplay of factors is beyond our natural capacity of analysis," explains Disc4All project coordinator Jérôme Noailly from Pompeu Fabra University in Barcelona, Spain.

"However, computer models and simulations can help us to retain only the most important cause to consequence relationships, providing us with sufficient understanding for well-informed predictions and actions."

Understanding processes triggering spine degeneration

Noailly's aim in this project was to provide a predictive mathematical tool to help spine care professionals make more informed decisions. However, he was finding it hard to make his models that depict what is going wrong mechanically in the spine correlate with what spine care professionals are trying to address and patients are actually feeling, which is pain.

The Disc4All project, supported by the Marie Skłodowska-Curie Actions programme, enabled Noailly to bring mathematical models closer to desired clinical support through the combination of AI and machine learning.

A core consortium of 12 beneficiaries brought together expertise in computer and data science, experimental and computational biology, bioinformatics, biomechanics and medicine.

This was achieved through integrating evidence-based data with biological and mechanical computational models.

The tools developed during Disc4All were applied to a range of data sources, which include both human cohorts and laboratory experiments. The aim was to provide a clearer picture of what is actually causing a patient intervertebral disc degeneration in the lower back, a major cause of lower back pain.

"We started looking more into the biological processes that actually trigger the degeneration of the spine," he adds. "By gathering data on this, I thought we might be able to identify personalised and enhanced descriptors of the causes of pain."

Grouping molecules according to function

AI and machine learning enabled Noailly and his colleagues to process this data rapidly, identifying objective risk factors of spine degradation along with demographic and psychological data.

"Medical image biomarkers were converted into personalised parameters in mathematical biomechanical models," Noailly explains. "This enabled us to compute and explore different biophysical mechanisms and the relation of each of these mechanisms with the personalised biomarkers."

Model simulations led to the identification of potential biomarkers, i.e. certain molecules that might be directly related to pain but not screened clinically. Noailly was able to see patterns in biological processes that are not easily captured in traditional ways.

"It's a little like how astronomers describe faraway phenomena without actually seeing them," he remarks. "They use complex models, and this is what we are doing here."

Meaning from biological and patient data

By applying machine learning to extract meaning from large amounts of biological and patient data, the project hopes to eventually help surgeons achieve a clearer picture of the causes of lower back pain in individual patients.

"AI-enhanced computer modelling is an efficient way of building up a personalised spinal model for precision medicine," says Noailly. "We are able to interpret data quickly and provide predictions as to whether personalised features represent a biological risk for intervertebral disc degeneration. This can then be leveraged to prevent or handle low back pain."

AI-enhanced computer modelling is an efficient way of building up a personalised spinal model for precision medicine.

The aim is to eventually bring this pioneering computational modelling concept into clinical settings. For healthcare providers, this could mean less reliance on expensive and time-consuming technologies such as CT and MRI scans.

PROJECT
Disc4All – Training network to advance integrated computational simulations in translational medicine, applied to intervertebral disc degeneration

COORDINATED BY
Pompeu Fabra University in Spain

FUNDED UNDER
Horizon 2020-MSCA-ITN

CORDIS FACTSHEET
cordis.europa.eu/project/id/955735

PROJECT WEBSITE
n/a

How ‘hearables’ track health by analysing bodily sounds

Taking the adage ‘listen to your body’ literally, researchers are developing AI-enabled wearable/mobile devices to track health by analysing our body’s sounds.



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Everyday wearable devices are proving a game changer for healthcare, with advanced sensors continuously monitoring aspects of wearers’ health. And as sensors drop in price, they will increasingly offer the kind of data-gathering previously only afforded by expensive medical equipment.

“This kind of longitudinal and fine-grained data is unprecedented in health tracking,” notes Cecilia Mascolo from the University of Cambridge and coordinator of the EAR project.

EAR was funded by the European Research Council.

EAR's researchers have developed a range of AI-enabled audio-based mobile health trackers, including earbuds that collect physiological information and a mobile phone app that monitors respiratory health.

Compared to AI for video and image analysis, its use for audio is underdeveloped, meaning the team had to develop novel workarounds.

"We are at the forefront of using AI to interpret audio data for health monitoring, using existing datasets and transfer learning to pretrain our models or existing pretrained models fine-tuned for our purposes using self-supervised learning," explains Mascolo.

Tracking general physiological health

Testing the hypothesis that the ear canal could prove an ideal location for a wearable health tracker, EAR developed an algorithm to repurpose earbud microphones to detect bodily sounds associated with health.

To test how well the earbuds could monitor users' gait and other physiological aspects, such as breathing and heart rate, volunteers participated in a variety of activities, including running and using rowing machines.

We are at the forefront of using AI to interpret audio data for health monitoring.

Drawing on other larger datasets, machine learning techniques trained an AI algorithm to identify and interpret the relevant health indicators in the audio.

"We have demonstrated that 'hearables' are good at collecting health-related audio which has indeed proven valuable for reflecting physiological health, thanks to our novel AI," adds Mascolo.

Tracking respiratory health

To help during the COVID-19 pandemic, the team wanted to see if a phone app could be developed to detect the presence of the coronavirus, using audio samples collected by the phone's microphone.

One of the largest multimodal audio datasets of oral sounds (breathing, coughing and speaking) was crowdsourced, combined with information about COVID-19 test status, symptoms and wider medical history.

This dataset is now informing machine learning models designed to forecast if an infected person will likely get better or worse, based on audio samples given regularly through the app.

Using the COVID-19 data, alongside other public health data, the team has also built the first audio-based respiratory model, OPERA, enabling researchers to perform tasks for which they had otherwise only limited data, such as assessing chronic obstructive pulmonary disease.

"Our COVID-19 data collection is unprecedented and remains in demand, with huge potential for this technology to track respiratory infections and chronic diseases over time," says Mascolo.

Longitudinal health monitoring at scale

EAR's innovative 'hearable' health solutions offer an affordable and scalable way to monitor individual and population health, significantly contributing to the EU's personalised and preventative health ambitions.

"We are now researching how audio in general, especially audio from the ear could yield additional health information as yet undetected or costly to detect otherwise," adds Mascolo.

In the meantime, a patent has been filed for the earbuds (for respiration rate monitoring), with Mascolo also joining a start-up working on in-ear microphone physiology monitoring through AI.

PROJECT
EAR – Audio-based Mobile Health Diagnostics

COORDINATED BY
University of Cambridge in the United Kingdom

FUNDED UNDER
Horizon 2020-ERC

CORDIS FACTSHEET
cordis.europa.eu/project/id/833296

PROJECT WEBSITE
health-sounds.cl.cam.ac.uk

Analysing evolving pollen exposure with AI

With climate change increasing the duration of the pollen season, AI is helping to reconstruct historic pollen exposures for research into health impacts.



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Pollen allergies (hay fever) are immune system reactions to airborne pollen grains released by trees, grasses and weeds, with common symptoms including sneezing, nasal congestion and watery, itchy eyes.

A recent analysis found that pollen allergies affect a median of 18.1 % of the global population, though estimates range widely between 1 % and 63 % for different countries.

Treatment typically involves medication such as antihistamine tablets, nose sprays and eye drops, alongside minimising exposure by staying indoors.

"We could also design public health interventions, like planting fewer allergenic trees in urban spaces, and combating climate change will help tackle invasive allergenic species like ragweed," says EPOCHAL project coordinator Marloes Eeftens from the Swiss Tropical and Public Health Institute. "But to monitor the success of these efforts, we first need to know more about changing pollen levels, and their health impacts."

AI-enabled pollen modelling

To assess how climate change has impacted pollen levels, EPOCHAL, funded by the European Research Council, used AI to analyse data from Switzerland's 14 pollen measurement

stations, considering 12 of the most allergenic species over 31 years.

"We found that for many allergenic plants in Switzerland, the pollen season starts earlier in the year, lasts longer and is more intense than before," notes Eeftens.

To explore exposure to pollen over time, EPOCHAL used AI to reconstruct daily pollen levels and distribution patterns for five types of pollen in Switzerland – Alder, Ash, Birch, Grass and Hazel – going back to 2002.

Modelling spatial and temporal variations

Machine learning tools were used to model spatial and temporal variations in pollen concentration, using pollen levels recorded by the measurement stations combined with predictor variables affecting emission, dispersion and sedimentation (e.g. altitude, land use, species distribution, temperature and vegetation).

To find the best performing machine learning method, six algorithms were compared. A learning method called Random Forest was found to be the best performing option.

"While we had known pollen levels at the measurement stations, exposure levels elsewhere previously had to be assumed. Our model was able to indicate the pollen types that homes throughout Switzerland were exposed to on specific days and in what concentrations over a 20-year period," adds Eeftens.

Effects of pollen on populations and individuals

EPOCHAL's dataset of historic pollen levels across Switzerland offered a means to compare exposure to health data, to study the effects on populations and individuals.

Some 410 volunteers were followed throughout the pollen season. The pollen they were exposed to was determined by EPOCHAL's models and their pollen sensitisation measured by skin prick tests, while symptom severity was self-reported.

This is all important information for pollen warning systems to communicate what people with allergies can expect.

Other health indicators including cognitive function, lung function, blood pressure and sleep quality during the pollen season were also measured.

"We found that there is no threshold below which symptoms can be completely avoided. Once exposure is above 50-80 pollen per cubic metre, suffering is typically moderate to severe, regardless of exact levels," explains Eeftens.

Analyses have already revealed for the first time, an association between pollen concentration and blood pressure, measured longitudinally.

Whereas, no significant relationship was found between pollen and cognitive performance.

"This is all important information for pollen warning systems to communicate what people with allergies can expect," adds Eeftens.

PROJECT
EPOCHAL – Beyond seasonal suffering: Effects of Pollen on Cardiorespiratory Health and Allergies

COORDINATED BY
Swiss Tropical and Public Health Institute in Switzerland

FUNDED UNDER
Horizon 2020-ERC

CORDIS FACTSHEET
cordis.europa.eu/project/id/853568

PROJECT WEBSITE
swisstph.ch/en

Efficient, accurate endometriosis diagnoses powered by AI

Clinical decision support tools powered by AI could help doctors identify endometriosis earlier, bringing significant health benefits to patients.



© Tom/stock.adobe.com

Endometriosis – a condition in which tissue similar to the lining of the uterus grows outside the uterus – affects around 190 million women worldwide, yet it often goes undiagnosed. Its symptoms are frequently dismissed, and diagnosis can take anywhere from 4 to 11 years.

The EU-funded FEMaLe project set out to address this challenge, by building clinical decision support tools powered by AI. The goal was to help doctors identify endometriosis earlier and with greater precision by analysing a wide range of patient data, from clinical records and self-reported symptoms to genetics and health histories.

Flagging early indicators of endometriosis

"We needed to make the invisible visible," explains Ulrik Bak Kirk, chief consultant at Aarhus University and FEMaLe project coordinator. "So, we brought technology to a space that has been neglected for decades."

At its core, FEMaLe used AI to analyse vast amounts of patient information – from clinical records and questionnaires to genetic data – to identify subtle patterns that often elude traditional diagnostics. Endometriosis isn't a one-size-fits-all disease. The symptoms vary widely, and the data is complex.

Rather than focusing on one symptom or test result, the AI looked at the full picture, learning from thousands of data points to flag early indicators of endometriosis.

"We applied AI and machine learning to identify early signs that may not have been immediately apparent," adds Kirk. "The earlier we can detect the disease, the better we can manage it – or even prevent it from progressing."

Empowering patients with data

The project developed two key systems: a clinical decision support system (CDSS) for healthcare professionals and a digital companion app for patients. Both were connected through a data infrastructure that allowed continuous feedback and iterative machine learning.

We brought technology to a space that has been neglected for decades.

FEMaLe's approach was holistic. It integrated patient voices directly into the development process and its pilots across Denmark, Hungary and Sweden – not just as subjects but as co-creators. Patient organisations and expert panels were part of the consortium from the start, helping ensure the tools were practical, ethical and grounded in lived experience.

"We engaged patients from day one," Kirk remarks. "Their feedback shaped the process every step of the way."

One of the project's core aims was to cut diagnostic delays. The CDSS supported doctors in assessing the likelihood of endometriosis by flagging combinations of symptoms and health histories that might otherwise be overlooked.

Meanwhile, the patient-facing digital companion app gave users more control over their condition. It allowed them to track symptoms, treatments and pain levels over time. This data was anonymised and then fed back into the AI engine, helping it to learn and improve diagnostic suggestions for future users.

Developing new health diagnostics

FEMaLe's influence has already extended beyond its original scope. Findings have helped develop new diagnostics such as the saliva-based Ziwig Endotest, while in Germany, projects MIRACUM and GECCO are also exploring integrated health data and AI-powered diagnostics, building on learnings from FEMaLe.

Now that the project has ended, the focus has turned to what happens next. Stakeholders are exploring how the tools developed can be integrated into national healthcare systems, expanded beyond Europe and adapted to other complex conditions with similar diagnostic challenges.

FEMaLe was a collaborative effort to ensure that women's pain was not just heard, but understood, measured and treated. "This isn't just about tech," Kirk concludes. "It's about trying to end the silence and stigma around a condition that affects so many millions of women."

PROJECT
FEMaLe – Finding Endometriosis using Machine Learning

COORDINATED BY
Aarhus University in Denmark

FUNDED UNDER
Horizon 2020-HEALTH

CORDIS FACTSHEET
cordis.europa.eu/project/id/101017562

PROJECT WEBSITE
findingendometriosis.eu

Boosting food safety in Europe with digital innovation

Digital technologies are helping the food safety industry turn data and information into action, right across the supply chain.



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In a world of increasingly complex and globalised food supply chains, ensuring food safety is a challenging task. Emerging hazards, from microbial outbreaks to chemical contamination, are placing constant stress on food systems.

In response, the EU-funded FoodSafeR project brought together 19 academic, research and industry organisations from across Europe to develop cutting-edge digital tools to address this challenge.

Harnessing AI for emerging risks

"The vision for FoodSafeR was to provide a proactive system for detecting and managing emerging food safety hazards," explains Lauren Alteio, a scientist at the Austrian Competence Centre for Feed and Food Quality, Safety & Innovation (FFoQSI). "We're not just monitoring risks though; we're trying to stay ahead of them."

AI was applied within research case studies, including the use of machine learning algorithms in bioinformatic approaches for microbiological use cases, and in the analysis of satellite imagery data for the investigation of mycotoxin contamination in crops.

In the age of 'big data', AI will become increasingly important to ensure a safer food future.

The project also applied machine learning and natural language processing to scan media and scientific data for early signs of emerging risks, in order to help professionals respond quickly and confidently.

Through this work, the project developed a framework for detecting how food safety threats change under pressure. "We identified indicators that allowed us to track how a threat behaves over time, essential for short-term response and long-term prevention," says Alteio.

These insights were then integrated into digital tools and policy recommendations for both operators and regulators. So far, four microbial and four chemical threat scenarios have been explored in depth, helping researchers understand how risks change and how to respond effectively.

The power of a DigitalHUB

Central to FoodSafeR was the creation of the DigitalHUB, a collaborative platform designed with and for food safety professionals. Developed through co-design events called Living Labs, the HUB serves as a digital meeting ground where users from academia, industry and government share data, tools and insights.

"We built the DigitalHUB with the people who will use it. It's designed to be flexible, transparent and easy to integrate into real-world food safety operations," notes Alteio.

Launched as an open access platform in 2025, the HUB offers tailored workspaces, networking features and document-sharing tools. It also includes training modules, a digital toolkit and AI-powered features such as Signal Trends.

Shaping the future of food safety

Dissemination was vital to the success of FoodSafeR, with the project sharing its outcomes through active participation in industry conferences such as FoodMicro 2024 and the IAFP European Symposium 2025, as well as hosting webinars on YouTube.

With its milestones achieved and the DigitalHUB now online, the project's next challenge is growing its user base and building a truly global community. Engagement and feedback will shape how the platform evolves and scales across the food safety landscape.

"In the age of 'big data', AI will become increasingly important and assist in informing decisions to ensure a safer food future," Alteio adds. Thanks to FoodSafeR, early detection, real-time data and AI-driven insights are transforming food safety management into a more adaptive, resilient system ready to tackle the threats of tomorrow.

PROJECT
FoodSafeR – A joined-up approach to the identification, assessment and management of emerging food safety hazards and associated risks

COORDINATED BY
Austrian Competence Centre for Feed and Food Quality, Safety & Innovation in Austria

FUNDED UNDER
Horizon Europe – Food, Bioeconomy, Natural Resources, Agriculture and Environment

CORDIS FACTSHEET
cordis.europa.eu/project/id/101060698

PROJECT WEBSITE
foodsafer.com

AI-powered platform to support healthy ageing

AI-powered healthcare platform helps prevent, manage and treat age-related conditions with personalised guidance, interventions and monitoring.



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Europe's ageing population has profound implications. Aside from the impact on individuals, increased chronic disease and long-term care needs place greater financial and emotional strain on families.

Additionally, while the state must meet the increasing cost, pension demands will also increase as the workforce contracts, potentially exacerbating labour shortages and reducing economic growth.

Against this backdrop, the EU-funded GATEKEEPER project has developed an AI-driven big data platform that offers next-generation healthcare services.

"Bringing together diverse stakeholders and 40 000 EU citizens, GATEKEEPER has helped pioneer innovative digital healthcare based on predictive risk assessments and personalised chronic disease management," explains Germán Gutiérrez, project coordinator, based at Medtronic in Spain.

Treatment of age-related conditions

To collect and aggregate data about patient health for the GATEKEEPER platform, including that from electronic medical records and personal health records, the team first developed data processing implementation guidelines based on a standard called HL7 – Fast Healthcare Interoperability Resources.

The platform's AI algorithms and models were then able to analyse and interpret this data to assess individual health risks, identify potential problems and provide personalised recommendations and interventions.

For instance, GATEKEEPER's algorithms were able to identify those most at risk of chronic conditions such as Type 2 diabetes. Through wearable devices such as smart watches, they were then able to offer personalised advice on the best fitness regimes to follow and even monitor adherence.

Piloting healthcare platform around the world

GATEKEEPER was piloted across various regions in Europe. In each case to evaluate the platform in real-world settings, the pilot team collaborated with healthcare providers and patients, alongside industry, civil society, academia and government stakeholders.



The pilots were successfully deployed across regions with very different healthcare needs and integrated into a range of healthcare settings, validating the platform's scalability and adaptability.

Guided by the European Innovation Partnership on Active and Healthy Ageing initiative, use cases included wearables which monitored glycaemic status in diabetes patients and stroke prevention using coaching apps and virtual reality.

"The pilots were successfully deployed across regions with very different healthcare needs and integrated into a range of healthcare settings, validating the platform's scalability and adaptability," notes Leandro Pecchia, pilot manager.

Moving forward, the team have streamlined an ethical and legal framework for healthcare innovation, established cybersecurity and data anonymisation protocols, and ensured compliance with data processing regulations, such as the GDPR.

Scaling up, rolling out and informing policy

Hewlett Packard's Ezmeral Platform secure cloud infrastructure currently hosts Gatekeeper, providing a secure environment for continued data storage and analysis. While a governance model is under development for the scaling up and rolling out of the system across various healthcare settings, pilot partners continue to advance GATEKEEPER's work.

For example, project partner Samsung has used project assets, such as its knowledge encoding model HeLiFit and an AI-enabled tool (MatKG) to create a knowledge graph from medical records and wearables data, which could be integrated into the Samsung Health App, to expand its functionality.

Also inspired by GATEKEEPER findings, Samsung has been developing and deploying AI-powered analysis to assess how well wearables track the impact of health interventions. Looking specifically at Type 2 diabetes and with a paper pending, the Samsung team found that wearables provided an unprecedented data richness correlating lifestyle (such as better sleeping and higher-intensity exercise) with HbA1c levels of glycated haemoglobin.

Again drawing on the work of GATEKEEPER, Samsung is also developing a Fast Healthcare Interoperability Resources data model to further optimise its Health App, by extending its ability to collect data on a broader range of health-related data.

"We have already made MatKG open access as a self-service solution for other developers in this space, while inspired by GATEKEEPER's technical results and following WHO health guidelines, we continue enhancing our health products," adds Carlo Allocca from Samsung and GATEKEEPER's AI expert.

PROJECT
GATEKEEPER – Smart living homes – whole interventions demonstrator for people at health and social risks

COORDINATED BY
Medtronic Iberica in Spain

FUNDED UNDER
Horizon 2020-HEALTH and Horizon 2020-LEIT-ICT

CORDIS FACTSHEET
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PROJECT WEBSITE
gatekeeper-project.eu

Biodiversity protection through innovative decision tools

New AI-driven tools have been developed to help policymakers better integrate biodiversity issues into the decision-making process.



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While society is heavily dependent on healthy, thriving ecosystems, biodiversity conservation and climate change mitigation are not always at the forefront of policy and decision-making.

"Biodiversity has certainly gained in visibility in recent years at all levels," notes Pierre Bonnet, researcher at the French Agricultural Research Centre for International Development (CIRAD) in Montpellier. "However, it is still often treated as a secondary consideration, especially when weighed against immediate economic or political concerns."

Indeed, one challenge is making biodiversity information usable at the scale and speed that decisions are made.

Creating biodiversity decision support applications

To address this, Bonnet and his team set out to create support applications designed to guide decision-making and safeguard biodiversity for the future, through the EU-funded GUARDEN project.

The GUARDEN system they developed draws on recent advances in citizen science, deep learning, Earth observation (EO) and hybrid modelling to offer accessible and usable information. The system is based on the observation that biodiversity concerns are not just limited to species conservation, but also underpin ecosystem services essential to agriculture, water regulation and climate change mitigation.

The DSAs are designed to support land-use planners, policymakers and other stakeholders in making biodiversity-conscious decisions.

GUARDEN decision support applications (DSAs) were developed, comprising multiple analytical components. These include species identification, habitat mapping and scenario modelling. These modules are interoperable and modular, meaning users can combine the tools needed for their unique decision-making and obtain tailored outputs.

"These DSAs are designed to support land-use planners, policymakers and other stakeholders in making biodiversity-conscious decisions," adds Bonnet.

The tools incorporate multiple technological layers, including EO, and state-of-the-art AI models for species identification – such as classification of plant communities from georeferenced images and the identification of bird species from audio recordings.

"These approaches allow for scalable, semi-automated biodiversity monitoring in both natural and human-dominated ecosystems," says Bonnet.

The DSAs are currently being tested in several sector-specific case studies, notably in the fields of urban planning and infrastructure development.

Implementing AI and Earth observation in decision-making

Among the most significant results is the demonstration that advanced technologies, such as AI and EO, can be implemented in ways that directly support biodiversity-aware decision-making.

Some of these tools, such as GeoPl@ntNet, have already demonstrated the potential for both large-scale and fine resolutions. Tools from the project are also available online, contributing to the European knowledge commons and enabling further innovations.

"Some of the project's tools have been tested outside the European context, for example in Madagascar, to confront our approaches with very different environmental, political and economic contexts," explains Bonnet.

Driving global research collaboration

The development of the tools also successfully brought together a large community across Europe. Several tools and services are being further refined and tested with European and international institutions, including with the Biodiversa+ network, the Joint Research Centre (JRC) and within the context of the World Flora Online consortium.

"These interactions are helping shape the system for new real-world operational use and policy integration," says Bonnet. "The system is being built not as a one-off prototype, but as a flexible, evolving infrastructure that can adapt to emerging needs."

PROJECT
GUARDEN – safeGUARDing biodivErsity aNd critical ecosystem services across sectors and scales

COORDINATED BY
French Agricultural Research Centre for International Development (CIRAD) in France

FUNDED UNDER
Horizon Europe – Food, Bioeconomy, Natural Resources, Agriculture and Environment

CORDIS FACTSHEET
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PROJECT WEBSITE
guarden.org

Interdisciplinary training for health promotion interventions

A new AI-driven approach to science training aims to bridge the gap between knowledge and action, especially when it comes to tackling public health challenges.



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Traditional, siloed approaches are no longer adequate for tackling major public health challenges of our time. A rise in chronic conditions, emerging mental health issues, pandemics and deep-rooted health inequities reflect not just medical problems, but complex societal challenges. This makes specialists combining varied skills and knowledge key to protecting human health into the future.

"What's needed is not only interdisciplinary expertise, but transdisciplinary and trans-sectoral collaboration," explains Sebastien Chastin, professor of Health Behaviour Dynamics at Glasgow Caledonian University and coordinator of the Health CASCADE project.

"That means going beyond bridging academic disciplines to actively working across institutional, professional and societal boundaries. This is the essence of co-creation."

To fulfil this need, the Health CASCADE project, undertaken with the support of the Marie Skłodowska-Curie Actions programme, sought to establish a European network to train the next generation of scientists. The aim was to give them cross-disciplinary scientific expertise, and equip them to collaborate across sectors such as healthcare, government and civil society.

"Health CASCADE reflects a necessary shift from top-down, expert-driven models to collaborative, systems-based solutions," says Chastin.

Central to the project was the integration of new digital technologies and AI to enhance and support the co-creation process. This was used for example to analyse complex qualitative data and to support decision-making.

We aimed to build a lasting infrastructure for transformative, co-created science in public health.

"We are particularly proud of having delivered dozens of co-creation projects across Europe, working directly with communities, health professionals, policymakers and other stakeholders to address real-world challenges," remarks Chastin. "These projects served as living laboratories – testing, refining and validating our methodologies in diverse contexts and with diverse populations."

Creating a long-lasting European training network

The Health CASCADE project was designed to deliver a world-class training programme for 15 doctoral candidates, equipping them with the skills needed to become leaders in the emerging field of co-creation in public health.

The training programme covered a broad range of topics, from qualitative and quantitative research methods, to research dissemination and communication, public engagement and project and data management.

"The training was responsive, and candidates were encouraged to co-shape the programme by suggesting topics and methods that best supported their needs," notes Chastin.

Beyond the training, candidates took part in international secondments with partner organisations to broaden their skills and build professional networks. The project also built a transdisciplinary ecosystem for co-creation science, with shared tools, open training resources and collaborative processes between institutions.

"We aimed to build a lasting infrastructure for transformative, co-created science in public health," adds Chastin.

Advancing the concept of co-creation

Health CASCADE has shown how the application of digital technologies can facilitate co-creation as a viable scientific methodology in public health. A robust scientific foundation for the theory, methods, technologies and evaluation strategies was established, which can now be built upon.

Establishing co-creation within European healthcare

The researchers hope that the work of Health CASCADE will help to mainstream co-creation as a core approach within European healthcare systems, as an evidence-based method to address complex health challenges.

The team is now working to set up a dedicated social enterprise focused on building capacity for co-creation across Europe and beyond. Research will continue into avenues such as hybrid collective intelligence and trans-sectoral co-creation, through new doctoral networks, including the Data2Action project.

"We see this as a long-term movement with a growing ecosystem of people, organisations and ideas committed to redefining how knowledge and solutions are created for the public good," says Chastin.

PROJECT
Health CASCADE – Evidence-based co-creation methodology for bridging the implementation gap between knowledge and action in health promotion

COORDINATED BY
Ghent University in Belgium

FUNDED UNDER
Horizon 2020-MSCA-ITN

CORDIS FACTSHEET
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PROJECT WEBSITE
healthcascade.eu

Preventing mosquito-borne diseases with AI and citizen science

Through combining citizen science and AI, the EU-funded H-MIP project is helping to inform public health prevention strategies and increase understanding of mosquito-borne diseases.



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Whether it be dengue, chikungunya, Zika or malaria, mosquito-borne diseases are on the rise. In fact, every year mosquitoes are responsible for nearly 700 million infections and over 1 million deaths worldwide.

With vaccines and cures generally lacking, new approaches to addressing the spread of mosquito-borne diseases are needed. For the H-MIP project, funded by the European Research Council, that means diving into the data.

According to the project, the good news is that very few species of mosquitoes actually transmit diseases. The bad news is that scientists don't know exactly which species those are.

To help fill this knowledge gap, the project is conducting widespread surveillance to accurately identify which mosquitoes carry what disease. "Mosquitoes are an excellent source of information about the people they bite, while people are an excellent source of information about the mosquitoes that bite them," says the project website.

To get that information, project researchers are using the ubiquitous mobile phone. "Mobile phone positioning can provide novel insights about human-mosquito biting networks," notes the project.

Calling all citizen scientists

The project has developed a mobile application that enables citizen scientists to collect data about mosquitoes and their bites. "People like to complain about mosquitos, and our Mosquito Alert app (website in Spanish) lets us listen and harness this energy through citizen science," adds the project.

Anyone at least 18 years old is invited to participate in the project. All you have to do is download the app and start reporting mosquitoes and their breeding sites when you find them and record where you have been bitten.

This information will allow researchers to gain knowledge about the spatial and temporal patterns of bites and, in the future, incorporate it into risk models together with the density of mosquitoes and people.

Beyond using the app, citizens can participate in the project by volunteering to be interviewed or have H-MIP researchers inspect their house for mosquitoes. You can also submit your DNA for analysis to see which mosquitoes have bitten you. For these options, the project asks that you submit this consent form, and all data is collected with fully informed consent and in line with EU data protection standards.

AI at the front lines of disease prevention

With this data in hand, the project plans to use AI algorithms to help predict the risk a specific mosquito species poses for transmitting disease.

"Knowing who mosquitoes bite or how many times they bite gives us an estimate of the interaction rate between humans and mosquitoes, which is one of the most important parameters in determining disease risk according to many mathematical models," says Federica Lucati, one of the project's researchers.

The project's work was tested in Africa as part of a campaign to track an invasive form of urban malaria, demonstrating the important role AI can play in disease research and prevention.

PROJECT
H-MIP – Human-Mosquito Interaction Project: Host-vector networks, mobility, and the socio-ecological context of mosquito-borne disease

COORDINATED BY
Pompeu Fabra University in Spain

FUNDED UNDER
Horizon 2020-ERC

CORDIS FACTSHEET
cordis.europa.eu/project/id/853271

PROJECT WEBSITE
h-mip.com/en/

Stopping marine pollution at the source

The EU-funded MAELSTROM project is using AI and other emerging technologies to prevent plastic waste from entering the ocean.



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There is an estimated 83 million tonnes of plastic waste in our oceans. Whether that waste be plastic bottles and straws or those insidious particles known as microplastics, they pose a significant threat to both marine life and human health.

The challenge is that removing the plastics already in the ocean is an arduous and costly task. That's why the EU-funded MAELSTROM project is focused on stopping this waste from getting to the sea in the first place.

"Our main goal is to reduce the impact of marine litter in coastal ecosystems by identifying accumulation hotspots and removing the existing litter from the coastal seabed and the water column of rivers before it reaches the sea," explains Fantina Madricardo, a researcher at the National Research Council – Institute of Marine Sciences (website in Italian), the project's coordinating partner.

Using AI to identify and collect plastics

To achieve this goal, the project implemented two innovative and environmentally sustainable technologies, both of which leverage the power of AI.

By involving local, national and international policymakers, enhancing ocean literacy and engaging directly with coastal communities, MAELSTROM has paved the way to the long-term adoption of our marine litter removal and recycling technologies.

One of those technologies is the Robotic Seabed Cleaning Platform.

This completely new solution uses an AI-driven robot to automatically identify and collect plastics from the seafloor of coastal environments.

"Not only can the platform selectively remove marine litter in coastal areas and shallow waters, its high selectivity minimises the environmental impact of the cleaning activities, while automated and robotised processes ensure its efficiency," says Madricardo.

The platform was recently tested in the Venice coastal area, where it successfully collected 2 240 kilograms of marine litter. This litter was then recycled and transformed into new materials that can be used to produce new circular economy products.

A sorting robot driven by artificial intelligence

The project also developed a sorting robot capable of segregating the collected plastic waste.

"This innovation takes all the collected plastics and automatically separates them according to polymer type, ensuring that they are correctly recycled and further advancing our goal of achieving a zero-waste marine litter value chain," adds Madricardo.

The AI-driven sorting robot was successfully tested by project partner Tecnalia in San Sebastian.

More than technology, project also actively engages with communities

Although AI has proven to be a powerful tool for removing plastics from coastal areas, technology alone is not enough. "The marine litter issue requires a multilevel, multidisciplinary effort that includes not only science and technology, but also policymaking and – most importantly – the participation of citizens," notes Madricardo.

That's why, in addition to the technology, the project focused on actively engaging with the public to raise awareness about marine litter and highlight how technology can be used to help address the problem. This work included educational opportunities and beach clean-up events.

"By involving local, national and international policymakers, enhancing ocean literacy and engaging directly with coastal communities, MAELSTROM has paved the way to the long-term adoption of our marine litter removal and recycling technologies," concludes Madricardo.

The project received an Atlantic Project Award in the Healthy Oceans and Resilient Coasts category and a Blue Rivers and Lakes Award. It was also recognised by the European Robotic Forums for sustainability leadership.

PROJECT
MAELSTROM – Smart technology for MArinE Litter SusTainable RemOval and Management

COORDINATED BY
National Research Council (CNR) in Italy

FUNDED UNDER
Horizon 2020-FOOD

CORDIS FACTSHEET
cordis.europa.eu/project/id/101000832

PROJECT WEBSITE
maelstrom-h2020.eu

Breakthrough biomarker discovery for dementia

Biomarker identification using AI tools could transform dementia care, enabling early diagnosis, precise monitoring and more effective future treatments.



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Dementia is a growing global challenge, with 40 million individuals affected worldwide, a number projected to double every 20 years. In Europe alone, 8.8 million people live with the condition.

Despite its profound personal and societal impacts, dementia remains difficult to diagnose in its early stages, as clinical symptoms typically emerge years after the onset of pathological changes, and there is considerable overlap in clinical symptoms between different causes of dementia.

Integrative approach towards biomarker identification

Undertaken with the support of the Marie Skłodowska-Curie Actions programme, the MIRIADE project aimed to accelerate the discovery of novel biomarkers for dementia.

"We established a unique research and training programme to enable precision health approaches for early diagnosis, disease monitoring and better treatment outcomes," explains project coordinator Charlotte Teunissen from Amsterdam University Medical Centre (UMC).

Breakthroughs with machine learning

A central achievement of the MIRIADE project lies in its integrative approach to data analysis and biomarker identification. The consortium unified omics data from cerebrospinal fluid (CSF) and blood samples and forwarded over 30 candidate biomarkers for development. It combined AI, big data analysis, cutting-edge assay technologies and entrepreneurial skills training, advancing both scientific discovery and clinical application.

The AI tools developed were designed to support the project's PhD students and researchers in narrowing down which proteins out of thousands were worth investigating. They drew on existing datasets, literature reviews and experimental data collected within the project itself.

In Luxembourg, for instance, researchers applied refined statistical models and text mining techniques. This multilayered data approach enabled machine learning models to predict which proteins were not only biologically relevant but also feasible to measure in CSF or blood.

Another powerful use of AI was in evaluating whether a given protein could be detected using antibodies – reagents that bind to target proteins. "Normally there is a bit of trial and error in this process," says Teunissen. "But the AI models helped predict whether an antibody-protein interaction was likely to work before we even tested it in the lab."

For three of the project's 15 PhD students, AI wasn't just a method of investigation, it was the main focus of their research. The benefits of this have extended well beyond the scope of MIRIADE. These models are now informing new projects across neurodegenerative research, offering a smarter way to target biomarkers that work in real-world clinical settings.

Biomarker validation and assay development

Thanks to this big data approach, remarkable progress was made in biomarker assay development and clinical validation. Using advanced protein technologies, MIRIADE successfully developed and clinically validated assays for 16 biomarkers.

Among the prioritised biomarkers were dopamine decarboxylase for dementia, aquaporin-4 for Alzheimer's disease and synaptic biomarkers SNAP-25 and VAMP-2 in blood for frontotemporal dementia and Alzheimer's disease.

Translating findings into clinical practice

The translation of MIRIADE's findings into clinical and diagnostic use is now well under way. Biomarkers such as dopamine decarboxylase are being implemented in initial clinical trials, and plasma pTau assays are undergoing prospective evaluation. For other biomarkers, retrospective and prospective validation studies are planned to refine their clinical application.

Furthermore, the project has created a comprehensive roadmap to guide future biomarker development. This framework includes best practices for integrating AI, big data, assay development and clinical validation, as well as approaches for fostering stakeholder collaboration to accelerate the translation of biomarkers into practice.

AI helped predict whether an antibody-protein interaction was likely to work before we even tested it in the lab.

Looking ahead, the project will continue its work by refining biomarker validation, scaling up assay production and seeking funding to advance its discoveries. MIRIADE is shaping a future where timely diagnosis and effective treatment significantly improve patient outcomes and reduce the socio-economic burden of dementia worldwide.

PROJECT
MIRIADE – Multi-omics Interdisciplinary Research Integration to Address DEmentia diagnosis

COORDINATED BY
Amsterdam University Medical Centre (UMC) in the Netherlands

FUNDED UNDER
Horizon 2020-MSCA-ITN

CORDIS FACTSHEET
cordis.europa.eu/project/id/860197

PROJECT WEBSITE
miriade.eu

Intelligent approaches to improved disease prevention

A powerful and freely available online platform that leverages data science and modelling can help to boost Europe's readiness against emerging infectious diseases.



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Climate change and the global movement of people and goods have increased the risk of disease threats. The COVID-19 pandemic also highlighted the need for improved surveillance and epidemic intelligence systems for the early detection, monitoring and assessment of emerging infectious diseases.

A key challenge is that current disease surveillance systems rely on routinely collected data, such as passive surveillance, to monitor disease occurrence and spread and to design appropriate responses. New approaches that incorporate AI, machine learning and the analysis of big data from a range of sources could help

improve preparedness by better understanding the drivers of disease emergence and enabling the development of more accurate models.

Responding to infectious disease threats

A good example of this is the EU-funded MOOD project. Through collecting and mining data using machine learning, this initiative has developed a unique digital platform designed to boost Europe's capacity to detect and respond to infectious disease threats through a 'One Health' approach.



Our aim was to bring complex data science and modelling tools directly into the hands of those doing research and risk assessment for science-based public health decisions.

"Our aim was to bring complex data science and modelling tools directly into the hands of those doing research and risk assessment for science-based public health decisions," explains project coordinator Elena Arsevska from the French Agricultural Research Centre for International Development (CIRAD) in France.

MOOD employed mathematical, statistical and data science-powered methods in the platform's infrastructure, and integrated machine learning algorithms to generate predictive risk maps for diseases.

In addition, the consortium incorporated the PADI-web tool, which uses natural language processing to automatically scan, extract and analyse disease-related information from online media to support event-based surveillance.

Data science for predictive modelling

The end result is an open access platform that brings together environmental, climatic, host distribution and disease data. The aim is to help public and animal health professionals make faster and better-informed decisions about emerging disease threats.

For example, the platform currently offers disease risk mapping outputs for various pathogens, including the West Nile virus, tick-borne encephalitis in humans, avian influenza in birds, and antimicrobial resistance in food-producing animals. "We will add more diseases to the platform in the future, depending on European needs," says Arsevska.

Over 20 partner institutions across Europe and the United States worked closely to ensure the platform met real-world, i.e. end user, needs. "MOOD's strength lies not just in the technology, but in its application," notes Arsevska.

Real-world impact on public health strategies

The project carried out specific case studies to demonstrate the platform's real-world impact on public health strategies. Each study incorporated data integration, predictive modelling and the 'One Health' perspective, reflecting the interdependence of human, animal and environmental health.

These case studies addressed infectious diseases representing different transmission routes, temporal dynamics and geographical distribution across Europe in the context of climate change.

For example, MOOD researchers showed that habitat, land-use and wildlife data can enhance local-level risk mapping in Europe for tick-borne encephalitis. In the case of the West Nile virus, the team combined indicator and event-based surveillance data to map the risk of disease occurrence and model the force of infection using longitudinal data from surveillance.

The avian influenza case study integrated traditional outbreak reporting with genomic data to assess transmission risks from wild birds to domestic poultry. This was also used to map hotspots for disease occurrence.

The case study on antimicrobial resistance revealed critical gaps in intersectoral data integration, demonstrating how resistant pathogens in livestock pose uneven risks across regions. The findings highlighted the urgent need for more coordinated 'One Health' surveillance strategies.

During the COVID-19 pandemic, MOOD works helped assess virus spread, evaluate the impact of mobility restrictions and inform interventions for public health agencies in Europe.

"MOOD helped future preparedness by strengthening collaborative platforms, research consortia and modelling networks to foster data and code sharing and effective collaboration between academia, decision makers and data providers," adds Arsevska.

To build on these results and secure sustainability beyond the project's lifetime, MOOD has established an international non-profit association to maintain and promote the platform, while forging new collaborations and exploring funding opportunities.

PROJECT
MOOD – MOnitoring Outbreak events for Disease surveillance in a data science context

COORDINATED BY
French Agricultural Research Centre for International Development (CIRAD) in France

FUNDED UNDER
Horizon 2020-HEALTH

CORDIS FACTSHEET
cordis.europa.eu/project/id/874850

PROJECT WEBSITE
mood-h2020.eu

Satellite data powers water use compliance monitoring

Integrating data on water usage to ensure regulatory compliance could help regions experiencing drought make optimal and fair use of this valuable resource.



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Climate change is causing disruptions to water cycle patterns, while growing demand from agriculture and other sectors is putting huge stress on this critical resource. Careful monitoring and management, especially in regions of scarcity, has never been more important.

The EU-funded WaterSENSE project sought to help water managers with a tool that combines Earth observation (EO) data with hydrological models, as well as local data. In doing so, the project team hoped to provide more accurate mapping of water availability and ensure compliance with water regulations.

Addressing water usage and compliance

"A key focus of the project was on Australia," explains WaterSENSE project manager Steven Wonink from eLEAF in the Netherlands. "The reasons for this are that the country has serious drought challenges, and it is also very advanced with regards to policy."

Wonink points out that the state of New South Wales (NSW) – a state 20 times larger than his native Netherlands – has legislation that requires farmers and other water users to buy licences. These quantify the amount of water they can take.

"The challenge is that it is incredibly difficult to enforce this legislation over such a large area," he says. "It is hard to track who is using the water, whether this is legal, and who owns which permit."

Applying EO and modelling data

To address this, a compliance monitoring tool was developed using EO data from Copernicus satellites, in situ measurements and advanced hydrological models. "Although AI was not at the heart of this project, we saw a growing focus on AI as the technology developed," adds Wonink. "For example, we used it in our data processing, to identify and mask cloud cover."

The tool was trialled in Australia's Murray-Darling Basin – a significant agricultural area – and demonstrated capability to estimate the amount of water being applied to individual fields. A great deal of complexity however involved matching EO and modelling data with regulatory compliance data.



Although AI was not at the heart of this project, we saw a growing focus on AI as the technology developed.

For this, the team worked together with a demonstration partner, the NSW Department of Climate Change, Energy, the Environment and Water. A database of licence users and holders – which were not always the same – was built and integrated with water usage data. The project also focused on measuring environmental flows to help authorities understand how much water is needed to maintain a natural ecosystem. Project partners also looked into developing specific services for cotton farmers.

Improving water use in the cotton industry

Following successful testing of the concept in Australia, the project team is continuing as a consortium to bring the technology closer to market. The aim is to expand and improve the toolbox and tailor it more to specific user needs.

"The work of WaterSENSE is being continued in a new EU-funded project called REINFORM," says Wonink. "In this project, we are specifically focusing on improving water use in the cotton industry. Our aim as a consortium is to be able to deliver a broad array of services to clients down the line. This technology can be applied anywhere in the world."

Moving forward, Wonink believes that the data analysis capabilities of AI will continue to be harnessed and grow in importance. "What we have seen in WaterSENSE is that one approach doesn't give you the complete picture," he notes. "There are multiple ways to estimate water use. AI can improve data quality and accuracy, in particular when it comes to analysing uncertainty."

PROJECT
WaterSENSE – Making SENSE of the Water value chain with Copernicus Earth Observation, models and in-situ data

COORDINATED BY
eLEAF in the Netherlands

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Through harnessing the power of gene editing, researchers are pioneering new medical and biotech breakthroughs. This CORDIS Results Pack showcases the scientific advancements made by eight EU-funded projects that are providing new insights into human history, biology and technology.



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